

Lesson 9

Advanced Programming II



From states in code to the board on Wi-Fi.

01 Revisit: Timer

02 Finite-state machine

03 enum & switch case

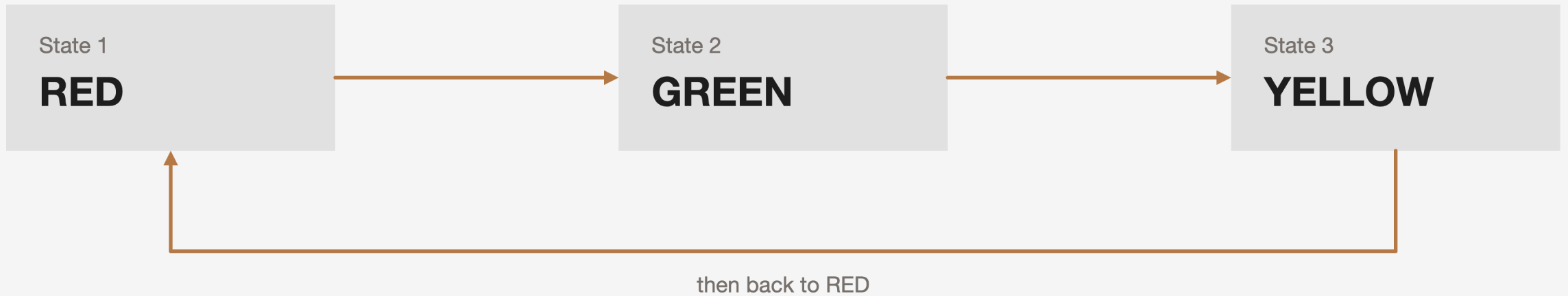
04 Wi-Fi & signal strength

Mark a moment on the timeline.

```
long last_tick = millis();
```

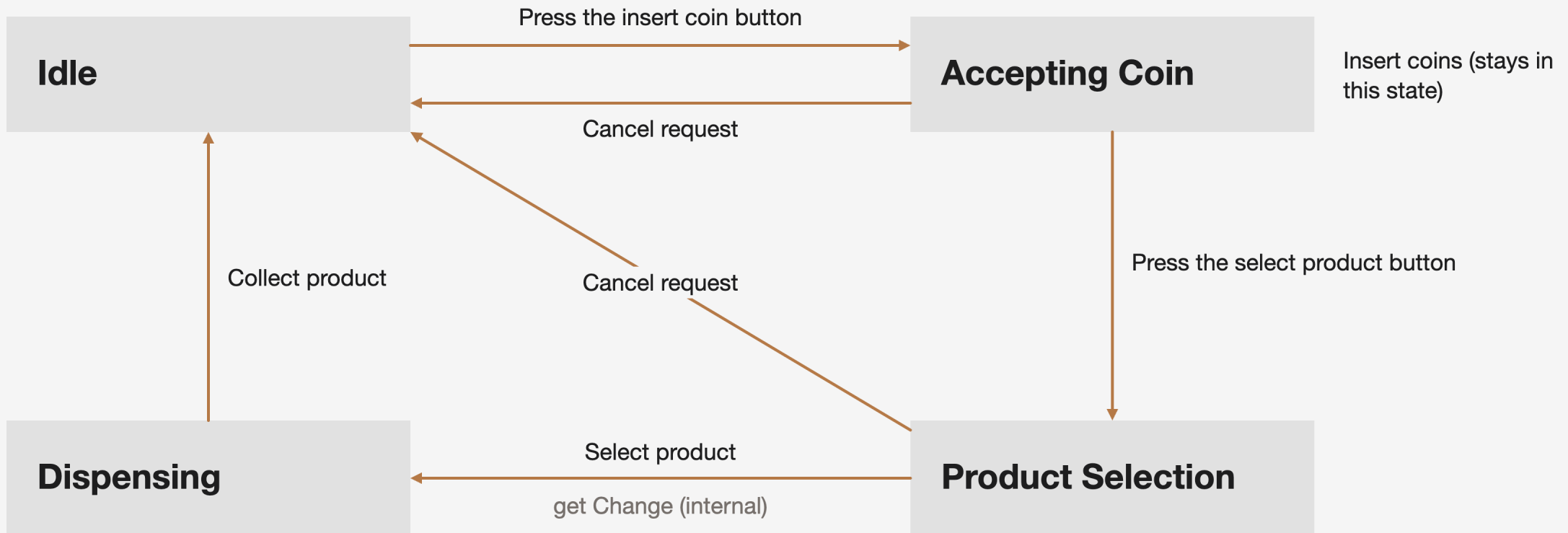


Traffic light: one state at a time.



The light is always in exactly one state, and it moves to the next state when its time is up.

Vending machine: four states.



Numbers can stand for names.

DAYS OF THE WEEK

Monday → 1

Tuesday → 2

...

Sunday → 7

TRAFFIC LIGHT STATES

RED → 1

YELLOW → 2

GREEN → 3

MOTIVATION

An **enum** is a way to **give friendly names** to a set of related integer constants, making code more readable. Less debugging time.

How to define your own enum.

TEMPLATE

```
enum EnumName  
{ VALUE1,  
  VALUE2,  
  VALUE3  
};
```

IN PRACTICE

```
enum TrafficLight  
{ RED,  
  YELLOW,  
  GREEN  
};
```

How to use your own enum.

DEFINE · NUMBERS START AT 0

```
enum TrafficLight
{ RED,           (0)
  YELLOW,        (1)
  GREEN,          (2)
  NUM_OF_STATES  (3)
};
```

DECLARE A VARIABLE

```
TrafficLight light = RED;
```

The enum name becomes a type, like int. light can only hold one of its values.

How to write switch case.

```
switch (light) {  
  case RED:  
    // turn on RED light  
    break;  
  case GREEN:  
    // turn on GREEN light  
    break;  
  case YELLOW:  
    // turn on YELLOW light  
    break;  
}
```

switch (light)

Checks the value of light.

case RED:

Runs when light is RED.

break;

Leaves the switch. Without it, the next case runs too.

All together.

1 · DEFINE

```
enum TrafficLight
{ RED,
  YELLOW,
  GREEN
};
```

2 · DECLARE

```
TrafficLight light = RED;
```

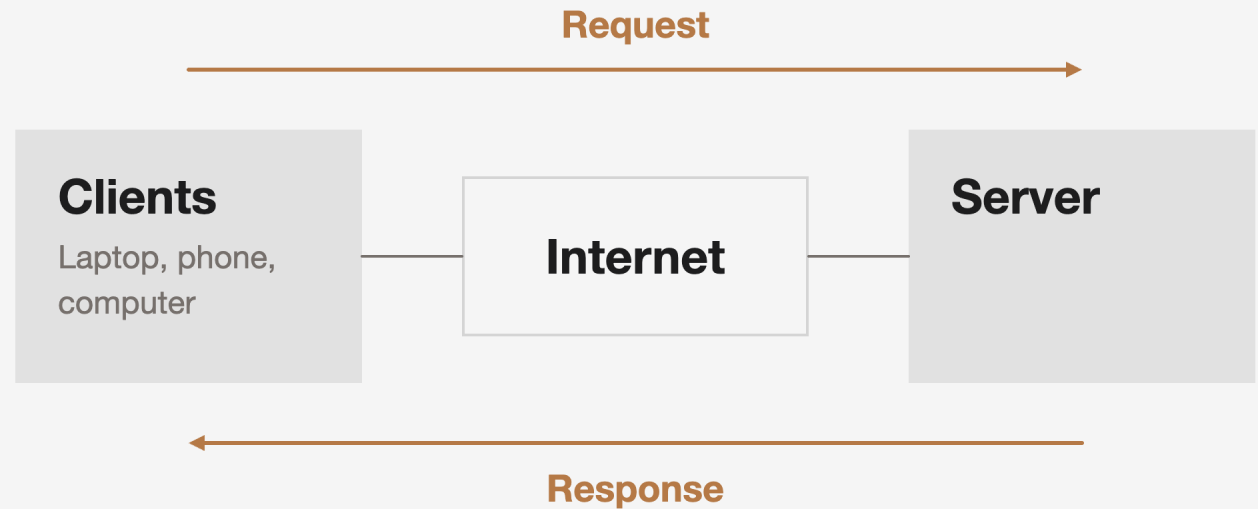


3 · SWITCH

```
switch (light) {
  case RED:
    // turn on RED light
    break;
  case GREEN:
    // turn on GREEN light
    break;
  case YELLOW:
    // turn on YELLOW light
    break;
}
```

Basic technical terms.

SSID	Network name
Password	Security key
IP address	Device identifier
Client vs server	Who asks, who answers



Connect the board to Wi-Fi.

BEFORE SETUP()

```
#include <WiFi.h>

const char* ssid =
    "YourWiFiName";
const char* password =
    "YourWiFiPassword";
```

LOOP()

```
void loop() {
    // Do nothing (or add more
    // functionality later)
}
```

SETUP()

```
void setup() {
    Serial.begin(115200);
    WiFi.begin(ssid, password);

    Serial.print("Connecting to WiFi...");
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }

    Serial.println("\nConnected! IP Address: ");
    Serial.println(WiFi.localIP());
}
```

Step-by-step Wi-Fi connection.

Step 1

WiFi.begin()

Connects to Wi-Fi.



Step 2

WiFi.status()

Checks connection status.



Step 3

WiFi.localIP()

If success, gets the assigned IP address.

Closer to 0 dBm means a stronger signal.

Signal strength	RSSI (dBm)	Meaning
Excellent	-30 to -70 dBm	Your connection is strong and you shouldn't have any issues.
Good	-71 to -80 dBm	Your connection is good enough for most activities.
Fair	-81 to -90 dBm	Your connection is fine for most tasks, but you may experience some issues.
Weak	Below -90 dBm	Your connection isn't strong enough for most activities.

Read the RSSI every two seconds.

```
void loop() {  
  // Check WiFi connection status  
  if (WiFi.status() == WL_CONNECTED) {  
    // Get the RSSI (Received Signal Strength Indicator)  
    long rssi = WiFi.RSSI();  
    Serial.print("Signal strength (RSSI): ");  
    Serial.print(rssi);  
    Serial.println(" dBm");  
  }  
  else {  
    Serial.println("WiFi disconnected");  
  }  
  delay(2000); // Wait 2 seconds before next reading  
}
```

TRY IT

Add this loop() to the Wi-Fi sketch.

Compare your reading with the table on the last slide.

Four things to take away.

- 1 A finite-state machine is always in one state and moves between states on events.
- 2 An enum gives friendly names to related integer constants, counting from 0.
- 3 switch case runs the code for the current value; break ends each case.
- 4 `WiFi.begin()`, `WiFi.status()` and `WiFi.localIP()` connect the board; `WiFi.RSSI()` reads signal strength.